

Localized Spectral Response

and the Full Subcritical Pressure Law

Moving zeros, a polyhedral cusp,
and finite-size scaling for digital products

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Abstract

The latest critical-pressure report in the ProveIt repository leaves the atomic-phase asymptotic unresolved for absolute-moment exponents $0 < s \leq 1/2$. We close that local gap and obtain a quantitative result throughout $0 < s < 1$. For the normalized digital mask in every integer base $b \geq 2$,

$$P_{b,s}(c) = (1-s) \log b + (b-1) \pi s \tan(\pi s/2) |c| + O_{b,s,\alpha}(|c|^{1+s-\alpha}), \quad 0 < \alpha < s.$$

The missing ingredient is a cancellation estimate for the perturbation paired with the atomic left eigenfunctional. This pairing is $O(|c|)$ on C^α , although its total-variation norm is of order $|c| \log(1/|c|)$ and the full operator changes only by $O(|c|^{s-\alpha})$.

A stronger multivariate theorem moves the $b-1$ mask zeros independently. With their displacements denoted by ϵ , the leading pressure response is exactly $\pi s \tan(\pi s/2) \|\epsilon\|_1$: the tangent cusp is polyhedral, not quadratic, and opposite zero motions do not cancel. We prove a weak distributional expansion, an explicit atomic spectral gap, the pressure–moment identification, a compact-parameter uniform version, and an operator finite-size law with an elementary amplitude. Every repository-specific ingredient is rederived. Reproducible numerical diagnostics and ten further research questions accompany the proofs.

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1 The question being resolved

1.1 A published regularity question and a precise repository gap

Generalized Thue–Morse products connect digital exponential sums, Riesz products, and thermodynamic formalism. Gohlke, Kesseböhmer, and Schindler develop their spectral and fractal theory and obtain an analytic phase-dependence result at a particular integer pressure order [1]. Gohlke, Lamprinakos, and Schmeling prove continuity in the singularity location for all nonnegative pressure orders. Following their Theorem 2.3, they ask about stronger regularity at general positive orders [2]. These statements do not themselves determine a local modulus or a one-sided derivative at the exceptional atomic phase.

The repository has recently isolated that phase in three reports on integer, supercritical fractional, and critical pressure. The last of these, *Critical Cusps in Digital-Product Pressure* [3], proves a subcritical formula for $1/2 < s < 1$ and explicitly asks whether it continues to $0 < s \leq 1/2$. Its scalar first-order coefficient is already valid below the cutoff. The obstruction is an error bound of the form $O(|c|^{2(s-\alpha)})$, obtained by squaring a strong operator norm. There is no $\alpha > 0$ making that error $o(|c|)$ when $s \leq 1/2$.

The present article resolves this exact obstruction. The improvement is not an extrapolation from a first-order calculation. We prove an $O(|c|)$ bound after pairing the perturbation with the left eigenfunctional, and combine it with the strong eigenfunction estimate. The resulting nonlinear error is

$$O(|c|) O(|c|^{s-\alpha}) = O(|c|^{1+s-\alpha}) = o(|c|)$$

for every $0 < s < 1$ and $0 < \alpha < s$.

Editorial note (ProveIt, 2026-09-29). This article answers the research question “The remaining subcritical interval” of [3] at the atomic phase, for every $0 < s \leq 1/2$, and on $1/2 < s < 1$ it sharpens that article’s $o(|c|)$ remainder to $O(|c|^{1+s-\alpha})$. Nonzero phases remain open. The article is filed in the repository beside its predecessor as `thue-morse/Thue_Morse_Subcritical_Pressure/`; the repository snapshot it was written against is [5]. The two tables of section 10 are the recorded output of a run with `-full`. Since 2026-09-29, `code/verify.py` writes to `recomputed/` unless `-output-dir` is given, and it refuses to rewrite `data/` without `-full`.

1.2 A broader result than a threshold extension

A common translation moves every zero by the same amount. Instead, we allow independent small displacements of all the mask zeros while keeping the modulus of the leading polynomial coefficient fixed. The main theorem shows that the first pressure increment is a constant multiple of the sum of the absolute displacements. Thus the method identifies a multivariate tangent geometry, in addition to removing the artificial threshold at $s = 1/2$.

The proof also distinguishes three different kinds of convergence: strong operator convergence, convergence after pairing with a regular test function, and total-variation convergence. Confusing them would lose exactly the cancellation that makes the result work.

1.3 What is, and is not, claimed

Claim class	Scope
Proved here	Full fixed-subcritical cusp; quantitative remainder; independent-zero ℓ^1 law; weak distributional response; compact-parameter uniformity; finite-size operator and moment scaling.
Reproved background	Atomic rank-one spectral gap; translated-zero coefficient; comparison of polynomial moments with cylinder pressure.
Not claimed	Global phase regularity; a two-point Lipschitz theorem on an entire phase neighborhood; a uniform limit as $s \downarrow 0$ or $s \uparrow 1$; a sharp next coefficient; formal verification; established worldwide priority.

The repository snapshot used for comparison is

458ccfb80b9940220819091da107e82bb970dcca.

The inspected critical report has Git blob c5491a93464668a8e20dd9afcd664a1c4b6cb81. Its discussion of the remaining subcritical interval and its first research question specify the gap addressed here. The current primary literature and targeted searches were also checked. This establishes a concrete local contribution relative to those sources, not a guarantee that no related formula exists elsewhere. No unverified repository theorem is used as an axiom in the proofs below.

2 Definitions and main results

2.1 Digital masks with independently moving zeros

Fix an integer $b \geq 2$. Write $T_b x = bx \pmod{1}$ on $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, and let

$$z_j = \frac{j}{b}, \quad 1 \leq j \leq b-1.$$

For a real displacement vector $\varepsilon = (\varepsilon_1, \dots, \varepsilon_{b-1})$, define

$$p_\varepsilon(z) = \frac{1}{b} \prod_{j=1}^{b-1} (z - e^{2\pi i(z_j + \varepsilon_j)}), \quad (2.1)$$

$$m_\varepsilon(x) = |p_\varepsilon(e^{2\pi i x})| = \frac{1}{b} \prod_{j=1}^{b-1} 2 |\sin \pi(x - z_j - \varepsilon_j)|. \quad (2.2)$$

Only sufficiently small displacements are considered, so the zeros remain separated. Put $\delta = \|\varepsilon\|_\infty$. At the origin of parameter space,

$$m_0(x) = \left| \frac{1}{b} \sum_{r=0}^{b-1} e^{2\pi i r x} \right| = \left| \frac{\sin \pi b x}{b \sin \pi x} \right|, \quad (2.3)$$

with removable values at integers. A common displacement $\varepsilon = c\mathbf{1}$ gives $m_\varepsilon(x) = m_0(x - c)$ exactly. For independent displacements, the mask need not be a normalized probability g -function. The digital-product pressure defined below nevertheless makes sense without that additional property.

For $0 < s < 1$, the transfer operator is

$$(\mathcal{L}_{s,\varepsilon} f)(x) = \sum_{k=0}^{b-1} m_\varepsilon\left(\frac{x+k}{b}\right)^s f\left(\frac{x+k}{b}\right). \quad (2.4)$$

There is *no factor* $1/b$ in this definition. The operator acts on $C^\alpha(\mathbb{T})$, with $\|f\|_\alpha = \|f\|_\infty + [f]_\alpha$, where the seminorm uses circle distance. Spectral statements use the complexification.

Let $N = b^n$ and let $\mathcal{D}_{b,n}$ be the N intervals of length $1/N$. Define

$$Q_{n,\varepsilon}(z) = \prod_{r=0}^{n-1} p_\varepsilon(z^{b^r}), \quad \deg Q_{n,\varepsilon} = N - 1, \quad (2.5)$$

$$F_{s,n,\varepsilon}(x) = |Q_{n,\varepsilon}(e^{2\pi i x})|^s, \quad M_{s,n}(\varepsilon) = \int_0^1 F_{s,n,\varepsilon}(x) dx, \quad (2.6)$$

$$Z_{s,n}(\varepsilon) = \sum_{I \in \mathcal{D}_{b,n}} \sup_{x \in I} F_{s,n,\varepsilon}(x), \quad P_{b,s}(\varepsilon) = \lim_{n \rightarrow \infty} \frac{1}{n} \log Z_{s,n}(\varepsilon). \quad (2.7)$$

Existence of this limit near $\varepsilon = 0$ is part of the proof. For common displacement we abbreviate $P_{b,s}(c) = P_{b,s}(c\mathbf{1})$.

The useful atomic quantities are

$$H(x) = \frac{|\sin \pi x|}{\pi}, \quad H_s = H^s, \quad \beta_s = b^{1-s}, \quad \nu_s(f) = \int_0^1 \frac{f(x)}{H_s(x)} dx, \quad \Pi_s = H_s \otimes \nu_s. \quad (2.8)$$

Since $s < 1$, ν_s is bounded even on C^0 , and $\nu_s(H_s) = 1$. Finally set

$$J_s = \pi s \tan\left(\frac{\pi s}{2}\right), \quad A_s = \frac{2}{\pi s} \tan\left(\frac{\pi s}{2}\right). \quad (2.9)$$

2.2 The full subcritical law and its multivariate strengthening

Theorem 2.1 (Polyhedral pressure cusp). *Let $b \geq 2$, $0 < s < 1$, and $0 < \alpha < s$. For all sufficiently small ε , the pressure exists and equals the logarithm of a real positive, simple dominant eigenvalue of $\mathcal{L}_{s,\varepsilon}$ on $C^\alpha(\mathbb{T})$. Moreover,*

$$P_{b,s}(\varepsilon) = (1-s) \log b + J_s \|\varepsilon\|_1 + O_{b,s,\alpha}(\delta^{1+s-\alpha}). \quad (2.10)$$

The statement is uniform for s in a compact subset $K \subset (0, 1)$ when $0 < \alpha < \inf K$ is fixed: the displacement neighborhood and implicit constant can then be chosen independently of $s \in K$.

Corollary 2.2 (The previously missing interval). *For every $0 < s < 1$,*

$$P_{b,s}(c) = (1-s) \log b + (b-1)J_s|c| + O_{b,s,\alpha}(|c|^{1+s-\alpha}). \quad (2.11)$$

In the binary convention $\psi_c(x) = \log \cos^2 \pi(x - c)$, the pressure order is $q = s/2$. Thus for all $0 < q < 1/2$,

$$P(q\psi_c) = (1-2q) \log 2 + 2\pi q \tan(\pi q)|c| + O_{q,\alpha}(|c|^{1+2q-\alpha}), \quad 0 < \alpha < 2q. \quad (2.12)$$

In particular, at the formerly excluded endpoint $q = 1/4$, the linear coefficient is $\pi/2$.

The sine convention in [2] has its atomic phase at $1/2$. Replacing its parameter by $c + 1/2$ gives the cosine convention above; there is no change in the pressure normalization. In the cosine convention the classical binary Thue–Morse phase is $c = 1/2$, not $c = 0$.

Corollary 2.3 (Pointwise geometry and one-sided derivatives). *The point $\varepsilon = 0$ is a strict local minimum of $P_{b,s}$. For every fixed nonzero real vector v ,*

$$\lim_{t \downarrow 0} \frac{P_{b,s}(tv) - P_{b,s}(0)}{t} = J_s \|v\|_1. \quad (2.13)$$

Consequently the pressure is not differentiable at the atomic point. For common phase displacement its right and left derivatives at zero are $(b-1)J_s$ and $-(b-1)J_s$, respectively. It has exact pointwise Hölder exponent one there.

Here pointwise Hölder exponent one means an optimal bound against the value at the distinguished point. It is not a claim about a uniform two-point Lipschitz seminorm throughout a neighborhood.

2.3 Finite-size scaling

Theorem 2.4 (Operator and moment scaling). *Fix b, s, α as in theorem 2.1. There are $C_T < \infty$ and $\rho \in (0, 1)$ such that, whenever δ is sufficiently small and $n\delta \leq T$,*

$$\left\| \beta_s^{-n} \mathcal{L}_{s,\epsilon}^n - e^{nJ_s \|\epsilon\|_1} \Pi_s \right\|_{C^\alpha(\mathbb{T}) \rightarrow C^\alpha(\mathbb{T})} \leq C_T (\delta^{s-\alpha} + \rho^n). \quad (2.14)$$

The analogous moment formula is

$$b^{sn} M_{s,n}(\epsilon) = A_s e^{nJ_s \|\epsilon\|_1} + O_T(\delta^{s-\alpha} + \rho^n). \quad (2.15)$$

In particular, if $\epsilon = h\nu$, $h \downarrow 0$, $n \rightarrow \infty$, and $nh \rightarrow \tau \in [0, \infty)$, then

$$b^{sn} M_{s,n}(h\nu) \longrightarrow A_s e^{\tau J_s \|\nu\|_1}. \quad (2.16)$$

Constants can be made uniform for $s \in K \Subset (0, 1)$ and bounded T , with fixed $0 < \alpha < \inf K$.

The condition $n \rightarrow \infty$ in the last statement matters when $\tau = 0$. At fixed n the initial rank-one relaxation has not vanished; the ρ^n term records that initial layer explicitly.

3 Why the strong norm alone gives the wrong threshold

The following elementary perturbation lemma isolates the mechanism. It is useful whenever a distinguished left eigenfunctional regularizes a perturbation more effectively than the ambient operator norm.

Lemma 3.1 (Strong-weak eigenvalue remainder). *Let X be a complex Banach space and $L_0 \in \mathcal{B}(X)$. Suppose β is an isolated algebraically simple eigenvalue with $L_0 h = \beta h$, $\nu L_0 = \beta \nu$, and $\nu(h) = 1$. For $L = L_0 + \Delta$ with $\|\Delta\|$ sufficiently small, let λ be the nearby eigenvalue. Then*

$$\lambda = \beta + \nu(\Delta h) + O(\|\nu \Delta\|_{X^*} \|\Delta\|_{X \rightarrow X}). \quad (3.1)$$

The constant depends on L_0, h, ν and an isolating contour, but not on the particular small perturbation.

Proof. Choose a circle Γ enclosing only β . The resolvent identity and the Neumann series imply that

$$\Pi_\Delta = \frac{1}{2\pi i} \int_\Gamma (z - L_0 - \Delta)^{-1} dz$$

is defined, has rank one, and satisfies $\|\Pi_\Delta - h \otimes \nu\| = O(\|\Delta\|)$. Indeed the inverse on Γ is obtained by multiplying the old resolvent by a uniformly convergent Neumann series. The same contour formula, or its homotopy in the perturbation parameter, preserves rank.

Normalize

$$h_\Delta = \frac{\Pi_\Delta h}{\nu(\Pi_\Delta h)}.$$

The denominator is $1 + O(\|\Delta\|)$, hence is nonzero for small perturbations; $\nu(h_\Delta) = 1$ and $\|h_\Delta - h\| = O(\|\Delta\|)$. Applying ν to $L h_\Delta = \lambda h_\Delta$ gives the exact identity

$$\lambda - \beta = \nu(\Delta h_\Delta) = \nu(\Delta h) + \nu(\Delta(h_\Delta - h)).$$

The last term has the bound in equation (3.1). □

The customary norm-square estimate follows by replacing $\|\nu\Delta\|$ by $\|\nu\| \|\Delta\|$. Here that replacement throws away useful information. We will instead show

$$\|\Delta\|_{C^\alpha(\mathbb{T}) \rightarrow C^\alpha(\mathbb{T})} = O(\delta^{s-\alpha}), \quad \|\nu_s \Delta\|_{(C^\alpha(\mathbb{T}))^*} = O(\delta). \quad (3.2)$$

Neither estimate alone supplies the theorem. Their product is the nonlinear remainder that works for the entire subcritical interval.

4 The atomic operator: an explicit spectral gap

4.1 Coboundary and quadrature

The atomic mask satisfies, away from removable exceptional points,

$$m_0(y)^s = \frac{H_s(by)}{b^s H_s(y)}. \quad (4.1)$$

Multiplication by H_s therefore conjugates the atomic operator to a uniform inverse-branch average, on the functions for which the quotient is meaningful. Direct substitution and a change of variables give

$$\mathcal{L}_{s,0} H_s = \beta_s H_s, \quad \nu_s \mathcal{L}_{s,0} = \beta_s \nu_s. \quad (4.2)$$

The quotient f/H_s can be singular at zero. We do not assume it lies in an unweighted Hölder space.

Lemma 4.1 (Weighted quadrature estimate). *For $0 < s < 1$ and $0 < \alpha < s$, with $N = b^n$,*

$$\|\beta_s^{-n} \mathcal{L}_{s,0}^n f - H_s \nu_s(f)\|_\infty \leq C_{b,s,\alpha} (N^{-\alpha} + N^{s-1}) \|f\|_\alpha. \quad (4.3)$$

Proof. For $0 < x < 1$, telescoping equation (4.1) gives

$$\beta_s^{-n} \mathcal{L}_{s,0}^n f(x) = H_s(x) \frac{1}{N} \sum_{k=0}^{N-1} \frac{f((x+k)/N)}{H_s((x+k)/N)}. \quad (4.4)$$

The case $N = 1$ is immediate after enlarging the constant. For $N \geq 2$, compare the sum with the integral of f/H_s by splitting off the two endpoint cells. On these cells $H_s(y) \asymp d(y, \mathbb{Z})^s$. The endpoint integrals are $O(N^{s-1} \|f\|_\infty)$ after multiplication by $H_s(x)$. The first sampled endpoint term is bounded by

$$\frac{H_s(x)}{N H_s(x/N)} \|f\|_\infty \leq C N^{s-1} \|f\|_\infty,$$

and the other endpoint is identical after reflection.

On the interior cells, replacing the sampled numerator by its value at a point of the cell costs at most

$$C N^{-\alpha} [f]_\alpha \int_{1/N}^{1-1/N} H_s(y)^{-1} dy \leq C N^{-\alpha} \|f\|_\alpha.$$

The variation of the denominator contributes at most a constant times

$$N^{-1} \|f\|_\infty \int_{1/N}^{1/2} y^{-s-1} dy \leq C N^{s-1} \|f\|_\infty.$$

Here one may first compare each cell to the minimum and maximum of the smooth denominator and then sum its variation; no derivative of f is used. These estimates prove the uniform bound for $0 < x < 1$. At $x = 0$, only the atomic fixed-point inverse branch survives at every iteration, and the normalized value is $N^{s-1} f(0)$. Since $H_s(0) = 0$, the same estimate holds there. $\square$

Proposition 4.2 (Rank-one exponential relaxation). *For every $0 < s < 1$ and $0 < \alpha < s$, there exist $C < \infty$ and $\rho_0 < 1$ such that*

$$\|\beta_s^{-n} \mathcal{L}_{s,0}^n - \Pi_s\|_{C^\alpha(\mathbb{T}) \rightarrow C^\alpha(\mathbb{T})} \leq C \rho_0^n. \quad (4.5)$$

In particular, β_s is an algebraically simple isolated dominant eigenvalue. The constants can be chosen uniformly on compact s -intervals whose lower endpoint exceeds α .

Proof. Write $B = \beta_s^{-1} \mathcal{L}_{s,0}$. The sup norms of $B^n 1$ are uniformly bounded by lemma 4.1. For each fixed m , pair corresponding inverse branches of T_b^m ; they contract circle distance by b^{-m} . The finite collection of branch weights is s -Hölder and hence α -Hölder. The difference of two output values therefore gives

$$[B^m f]_\alpha \leq C_0 b^{-m\alpha} [f]_\alpha + C_m \|f\|_\infty, \quad (4.6)$$

where C_0 is independent of m and C_m is finite. The first coefficient comes from the sum of the nonnegative branch weights, namely $B^m 1$; the second comes from their Hölder seminorms. Across the endpoint of the circle the inverse branches are relabeled cyclically, giving the same estimate.

The subspace $\ker \nu_s$ is invariant. On it lemma 4.1 implies

$$\|B^n f\|_\infty \leq C \theta^n \|f\|_\alpha, \quad \theta = \max\{b^{-\alpha}, b^{s-1}\} < 1.$$

Choose m with $a = C_0 b^{-m\alpha} < 1$. If $u_k = [B^{km} f]_\alpha$, equation (4.6) yields

$$u_{k+1} \leq a u_k + C_m C \theta^{km} \|f\|_\alpha.$$

A finite geometric convolution bounds u_k by $C r^k \|f\|_\alpha$ for any fixed r strictly between $\max\{a, \theta^m\}$ and one. The finitely many remaining residues of n modulo m change only the constant. This proves exponential contraction on $\ker \nu_s$ in the full norm. Since Π_s is a commuting rank-one projection by equation (4.2), equation (4.5) follows.

An additional eigenvalue of modulus at least one for B , or a nontrivial Jordan vector at one, contradicts this exponential convergence. The spectrum on the kernel is contained in a smaller disk by the spectral radius formula, proving the stated spectral conclusions. All estimates are uniform when s stays away from zero and one and α is fixed below its lower endpoint; the same m may then be used. $\square$

5 Translated zeros as weak distributions

5.1 The universal one-zero constant

The following integral is the local contribution of opening a simple zero. Its evaluation appears in the predecessor report; we include the proof to keep the new argument independent.

Lemma 5.1 (Model singular integral). *For $0 < s < 1$,*

$$\int_0^\infty u^{-s} ((u+1)^s + |u-1|^s - 2u^s) du = J_s = \pi s \tan(\pi s/2). \quad (5.1)$$

The paired integrand is integrable at both zero and infinity.

Proof. At zero it is $O(u^{-s})$; at infinity the two linear Taylor terms cancel, leaving $O(u^{-2})$. Substitute $u = 1/v$ and integrate by parts. The endpoint terms vanish and the result, divided by s , is

$$\int_0^\infty \frac{(1+v)^{s-1} + \operatorname{sgn}(v-1)|v-1|^{s-1}}{v} dv. \quad (5.2)$$

The terms at zero must remain paired. Transform the first term using $t = v/(1+v)$ and pair it with the negative term on $(0, 1)$. Their sum is

$$\int_0^1 \frac{(1-t)^{-s} - (1-t)^{s-1}}{t} dt = \psi(s) - \psi(1-s) = -\pi \cot(\pi s).$$

The remaining term is

$$\int_1^\infty \frac{(v-1)^{s-1}}{v} dv = B(s, 1-s) = \pi \csc(\pi s).$$

The beta and digamma reflection identities used here are standard [4, §5.5]. Their sum is $\pi(\csc \pi s - \cot \pi s) = \pi \tan(\pi s/2)$. $\square$

5.2 A quantitative distributional expansion

For a point $z \in \mathbb{T}$, define

$$q_{z,h}(x) = \left| \frac{\sin \pi(x-z-h)}{\sin \pi(x-z)} \right|^s - 1, \quad k_z(x) = \pi \cot \pi(x-z).$$

Values at the zero denominator are irrelevant for integration. The notation PV always means deletion of a symmetric interval around that zero, followed by a limit.

Lemma 5.2 (One-zero weak derivative). *Let $0 < s < 1$ and $0 < \alpha < 1$. For $g \in C^\alpha(\mathbb{T})$,*

$$\int_0^1 q_{z,h}(x)g(x) dx = J_s|h|g(z) - sh \text{PV} \int_0^1 k_z(x)g(x) dx + O_{s,\alpha}(|h|^{1+\alpha} \|g\|_\alpha). \quad (5.3)$$

The principal-value functional is bounded on C^α . In particular, the left side is $O(|h| \|g\|_\alpha)$. For $g = 1$ the error can be improved to $O_s(h^2)$.

Proof. Translation lets us take $z = 0$. Choose a fixed small symmetric interval $(-r, r)$ about zero and use its signed real coordinate. There, $|\sin \pi x| = |x|a(x)$ with a positive and smooth. Set

$$U_h(x) = |x-h|^s|x|^{-s}, \quad A(x) = a'(x)/a(x).$$

Uniformly on this fixed interval,

$$q_{0,h}(x) = U_h(x) - 1 - shA(x)U_h(x) + O(h^2U_h(x)). \quad (5.4)$$

The integrals of U_h on $(-r, r)$ are uniformly bounded.

First take a constant test. Pairing positive and negative x and scaling by $|h|$ gives

$$\int_{-r}^r (U_h - 1) dx = |h|J_s + O(h^2),$$

because the tail of the paired integral in equation (5.1) is $O(1/R)$. The smooth correction in equation (5.4) contributes $-sh \int_{-r}^r A + O(h^2)$. To see the stated error, use the estimate $\int (U_h - 1)A = O(|h|)$: subtract $A(0)$, use its Lipschitz modulus, and pair the constant part. On the complement, ordinary Taylor expansion contributes $-sh \int k_0 + O(h^2)$. The local and complementary linear terms add to $-sh \text{PV} \int k_0 = 0$ by periodicity. This proves the constant-test assertion.

For a general test, subtract $g(0)$ in the local interval. On $|x| \leq 2|h|$, its modulus is at most $[g]_\alpha |x|^\alpha$. The integrals of

$$(U_h + 1)|x|^\alpha \quad \text{and} \quad |h||x|^{\alpha-1}$$

over that region are $O(|h|^{1+\alpha})$; integrability of the first uses $s < 1$ (indeed $\alpha - s > -1$). On $2|h| < |x| < r$, Taylor expansion gives

$$U_h - 1 + \frac{sh}{x} = O(h^2/x^2).$$

Multiplying by $g(x) - g(0)$ and integrating costs

$$Ch^2[g]_\alpha \int_{2|h|}^r x^{\alpha-2} dx \leq C|h|^{1+\alpha} \|g\|_\alpha.$$

The smooth factors in equation (5.4) and the complement have the same or a smaller remainder. One may use the elementary estimate $\int_{-r}^r |U_h - 1| dx = O(|h| \log(1/|h|))$; multiplying by $|h|$ is still $O(|h|^{1+\alpha})$ since $\alpha < 1$.

Finally, $\text{PV} \int_{-r}^r g(x)/x = \int_{-r}^r (g(x) - g(0))/x$ is absolutely convergent and bounded by $C[g]_\alpha$. The remaining smooth part is bounded by $C\|g\|_\infty$. This proves both the expansion and the boundedness of its principal-value term. $\square$

In distribution notation, for $h \downarrow 0$,

$$h^{-1}q_{z,\pm h}(x) dx \longrightarrow J_s \delta_z \mp s \text{PV}(k_z(x) dx) \quad \text{in } (C^\alpha)^*. \quad (5.5)$$

The Dirac term is even in the displacement; the principal-value term is odd. The Dirac term cannot be recovered by pointwise differentiation away from the zero.

5.3 Several zeros and locally regular test functions

Let

$$R_\epsilon(x) = \left(\frac{m_\epsilon(x)}{m_0(x)} \right)^s = \prod_{j=1}^{b-1} (1 + q_{z_j, \epsilon_j}(x)). \quad (5.6)$$

Fix disjoint closed neighborhoods U_j of the z_j , none containing zero. It is useful to allow a test function that is only integrable away from these neighborhoods. Write

$$\|g\|_* = \|g\|_{L^1(\mathbb{T})} + \sum_{j=1}^{b-1} \|g|_{U_j}\|_{C^\alpha(U_j)}.$$

Shrinking a neighborhood slightly when multiplying by a fixed cutoff only changes constants.

Proposition 5.3 (Multizero weak expansion). *For $0 < s < 1$, $0 < \alpha < 1$, and tests with finite $\|g\|_*$,*

$$\int_0^1 (R_\epsilon - 1)g dx = J_s \sum_{j=1}^{b-1} |\epsilon_j| g(z_j) - s \sum_{j=1}^{b-1} \epsilon_j \text{PV} \int_0^1 k_{z_j}(x) g(x) dx + O(\delta^{1+\alpha} \|g\|_*). \quad (5.7)$$

In particular this is $O(\delta \|g\|_)$. For $g = 1$,*

$$\int_0^1 (R_\epsilon - 1) dx = J_s \|\epsilon\|_1 + O(\delta^2). \quad (5.8)$$

Constants are uniform when s ranges over a compact subset of $(0, 1)$.

Proof. Use a smooth partition of unity subordinate to the U_j and their complement. On a neighborhood of z_j , every factor with index $k \neq j$ has a smooth expansion

$$q_{z_k, \epsilon_k} = -s\epsilon_k k_{z_k} + O(\delta^2)$$

with uniform bounds on a fixed number of derivatives. On the complement all factors have such expansions, and the error is bounded pointwise by $C\delta^2$, which can be integrated against $|g|$.

For the terms linear in one q_j , apply lemma 5.2 locally, using cutoffs, and use Taylor expansion elsewhere. A principal value at z_j is well defined because g is Hölder there; elsewhere its kernel is bounded against the integrable test.

It remains to control mixed products of two or more distinct q_j . Near z_j , a mixed term containing q_j has the form $q_j ag$, where a is smooth with C^1 norm $O(\delta)$, or smaller. Apply the $O(|\varepsilon_j|)$ functional bound from lemma 5.2 to the cutoff test ag . The result is $O(\delta^2 \|g\|_*)$. A mixed term not containing that local singular factor is pointwise $O(\delta^2)$. Thus all mixed terms are $O(\delta^2 \|g\|_*)$, not $O(\delta^2 \log(1/\delta))$. This signed estimate is the place where absolute-value estimates would be needlessly wasteful.

For $g = 1$, each unmixed term has the improved $O(\varepsilon_j^2)$ remainder from lemma 5.2, and PV $\int_0^1 k_{z_j} = 0$. Mixed terms retain the $O(\delta^2)$ bound. This proves equation (5.8). Separation of the fixed zeros and compactness of the exponent interval give the stated uniformity. $\square$

6 The localized left-eigenfunctional estimate

6.1 An exact change-of-variables identity

Let $\Delta_\varepsilon = \mathcal{L}_{s,\varepsilon} - \mathcal{L}_{s,0}$. Combining the change of variables $x = T_b y$ with equation (4.1) gives

$$\nu_s(\Delta_\varepsilon f) = \beta_s \int_0^1 (R_\varepsilon(y) - 1) \frac{f(y)}{H_s(y)} dy. \quad (6.1)$$

This identity is valid for every continuous f . Near a mask zero, the ratio has a $|y - z_j|^{-s}$ singularity; near the atomic point zero, H_s^{-1} is integrable and the ratio is smooth. Hence all its integrals exist when $s < 1$.

Define bounded functionals on $C^\alpha(\mathbb{T})$ by

$$E_j(f) = \frac{f(z_j)}{H_s(z_j)}, \quad D_j(f) = \text{PV} \int_0^1 k_{z_j}(y) \frac{f(y)}{H_s(y)} dy. \quad (6.2)$$

Near z_j the denominator is smooth and nonzero. Near zero the cotangent kernel is bounded, so the remaining integrability follows from $s < 1$. Thus D_j is well defined even though f/H_s need not be a globally continuous function.

Theorem 6.1 (Weak operator expansion). *For $0 < s < 1$ and $0 < \alpha < s$,*

$$\nu_s \Delta_\varepsilon = \beta_s \left(J_s \sum_j |\varepsilon_j| E_j - s \sum_j \varepsilon_j D_j \right) + O_{(C^\alpha(\mathbb{T}))^*}(\delta^{1+\alpha}). \quad (6.3)$$

Consequently,

$$\|\nu_s \Delta_\varepsilon\|_{(C^\alpha(\mathbb{T}))^*} \leq C\delta, \quad (6.4)$$

$$\nu_s(\Delta_\varepsilon H_s) = \beta_s J_s \|\varepsilon\|_1 + O(\delta^2). \quad (6.5)$$

Proof. For $g = f/H_s$ in proposition 5.3,

$$\|g\|_{L^1} \leq \|f\|_\infty \int H_s^{-1}, \quad \|g|v_j\|_{C^\alpha} \leq C \|f\|_\alpha.$$

Thus $\|g\|_* \leq C \|f\|_\alpha$, and equation (5.7) applied to equation (6.1) proves equation (6.3). For $f = H_s$ the quotient test is exactly one. Therefore every E_j equals one, every D_j equals zero, and the improved scalar estimate equation (5.8) proves equation (6.5). $\square$

6.2 Why a total-variation bound cannot replace cancellation

Proposition 6.2 (Sharp logarithmic loss in the continuous dual norm). *For common displacement $\varepsilon = c\mathbf{1}$ and fixed $0 < s < 1$,*

$$\|\nu_s \Delta_{c\mathbf{1}}\|_{(C^0)^*} = 2\beta_s s \left(\sum_{j=1}^{b-1} H_s(z_j)^{-1} \right) |c| \log \frac{1}{|c|} + O(|c|). \quad (6.6)$$

In particular, an $O(|c|)$ estimate in this larger dual norm is false.

Proof. By equation (6.1), the functional is represented by the signed integrable density $\beta_s(R_{c\mathbf{1}} - 1)H_s^{-1}$. Its norm on C^0 is its total variation; continuous tests approximate its sign in the corresponding L^1 space. In a signed coordinate u around z_j , for $2|c| < |u| < r$,

$$R_{c\mathbf{1}}(z_j + u) - 1 = -\frac{sc}{u} + O(c^2/u^2) + O(|c|).$$

Also $H_s(z_j + u)^{-1} = H_s(z_j)^{-1} + O(|u|)$. The two sides of the puncture therefore contribute $2sH_s(z_j)^{-1}|c| \log(1/|c|) + O(|c|)$. The inner interval $|u| \leq 2|c|$ contributes $O(|c|)$ by scaling and $s < 1$. Away from all the zeros the ratio difference is $O(|c|)$, and the atomic weight remains integrable. Summing and multiplying by β_s gives equation (6.6). $\square$

Thus the better estimate is genuinely weak: a Hölder test cannot follow the rapidly changing sign around a translated zero at unit amplitude. Pairing the two sides cancels the logarithmic divergence. The atom in equation (5.5) survives that cancellation and produces the cusp coefficient.

7 Strong perturbation and the pressure theorem

7.1 Translation in Hölder norm

Lemma 7.1 (Strong operator modulus). *For $0 < s < 1$, $0 < \alpha < s$, and sufficiently small ε ,*

$$\|\mathcal{L}_{s,\varepsilon} - \mathcal{L}_{s,0}\|_{C^\alpha(\mathbb{T}) \rightarrow C^\alpha(\mathbb{T})} \leq C\delta^{s-\alpha}. \quad (7.1)$$

Proof. Each function $u_j(x) = |\sin \pi(x - z_j)|^s$ is s -Hölder. Its translation difference satisfies

$$\|u_j(\cdot - h) - u_j\|_\infty \leq C|h|^s, \quad [u_j(\cdot - h) - u_j]_s \leq C.$$

Splitting the Hölder quotient at distance $|h|$, or using the minimum of these two estimates, gives

$$\|u_j(\cdot - h) - u_j\|_{C^\alpha} \leq C|h|^{s-\alpha}.$$

The identity $m_\varepsilon^s = b^{-s}2^{s(b-1)} \prod_j u_j(\cdot - \varepsilon_j)$, a telescoping product, and the Banach algebra estimate for C^α show that the weight difference has this same bound with $|h|$ replaced by δ . The finite sum of contracted inverse-branch compositions in equation (2.4) is bounded on C^α . Cyclic relabeling at the circle endpoint again makes this a global estimate. $\square$

7.2 The eigenvalue expansion

Proposition 7.2 (Quantitative spectral response). *Under the hypotheses of theorem 2.1, the eigenvalue near β_s is real, positive, and satisfies*

$$\lambda_s(\varepsilon) = \beta_s(1 + J_s \|\varepsilon\|_1) + O(\delta^{1+s-\alpha}). \quad (7.2)$$

It is simple and strictly larger in modulus than all other spectrum. If $\Pi_{s,\varepsilon}$ is its spectral projection, then

$$\|\Pi_{s,\varepsilon} - \Pi_s\| = O(\delta^{s-\alpha}). \quad (7.3)$$

Proof. Apply lemma 3.1 using proposition 4.2, theorem 6.1, and lemma 7.1. The linear scalar term is equation (6.5); the nonlinear error is $O(\delta^{1+s-\alpha})$. The scalar $O(\delta^2)$ remainder is smaller, since $s - \alpha < 1$. This proves equation (7.2) and the contour construction gives equation (7.3).

The atomic spectral gap supplies a fixed separating annulus. A Neumann-series resolvent argument on its compact boundary, together with the usual large- z resolvent bound, keeps the complementary spectrum in a disk of radius strictly below β_s for all small perturbations. The nearby eigenvalue tends to β_s and stays outside that disk. Complex conjugation preserves the operator and the isolating circle, so uniqueness of the eigenvalue in the circle makes it real; it is positive by its limit. These conclusions do not rely on a positivity theorem for a strictly positive potential. $\square$

7.3 Why cylinder pressure equals moment growth

This step is needed because a singular logarithmic potential should not be placed into a classical positive-potential theorem without checking its hypotheses.

Lemma 7.3 (Polynomial sampling bound). *For every fixed $s > 0$ there is $C_s < \infty$ such that, for every polynomial p of degree at most $N - 1$,*

$$N \int_0^1 |p(e^{2\pi i x})|^s dx \leq \sum_{I \in \mathcal{D}_N} \sup_{x \in I} |p(e^{2\pi i x})|^s \leq C_s N \int_0^1 |p(e^{2\pi i x})|^s dx. \quad (7.4)$$

Here $\mathcal{D}_N$ is the uniform partition into N intervals.

Proof. The lower bound is immediate by integration on each cell. For the upper bound choose $R = 1 + 1/N$. Subharmonicity of $|p|^s$ and the Poisson majorant on the disk of radius R give

$$|p(e^{2\pi i x})|^s \leq \int_0^1 P_{1/R}(2\pi(x - y)) |p(Re^{2\pi i y})|^s dy.$$

The Poisson kernel is bounded by $CN/(1 + N^2 \text{dist}(x - y, \mathbb{Z})^2)$. For fixed y , summing its supremum over the N cells gives at most CN : the bounded number of nearest cells contribute $O(N)$ and the others contribute a convergent $N \sum_{k \geq 1} k^{-2}$ tail. Consequently the cylinder sum is bounded by $CN \int_0^1 |p(Re^{2\pi i y})|^s dy$.

If $D = \deg p$, the reciprocal polynomial $q(z) = z^D p(1/z)$ and the monotonicity of radial means of a subharmonic function show that

$$\int_0^1 |p(Re^{2\pi i y})|^s dy \leq R^{Ds} \int_0^1 |p(e^{2\pi i y})|^s dy.$$

Indeed evaluate the radial mean of $|q|^s$ at radius $1/R$ and compare it with radius one. Since $D \leq N - 1$, $R^{Ds} \leq e^s$. This proves the upper bound. Constant and zero polynomials are covered by the same inequality directly. $\square$

A change of variables in the transfer operator gives the exact identity

$$\int_0^1 \mathcal{L}_{s,\varepsilon}^n 1 dx = b^n M_{s,n}(\varepsilon). \quad (7.5)$$

The spectral projection amplitude at the atomic point is

$$\int_0^1 \Pi_s 1 dx = \left(\int_0^1 H_s \right) \left(\int_0^1 H_s^{-1} \right) > 0. \quad (7.6)$$

By equation (7.3) this amplitude remains positive for small real ε . The spectral decomposition therefore gives the logarithmic growth rate $\log \lambda_s(\varepsilon)$ to the left side of equation (7.5). Applying lemma 7.3 to $Q_{n,\varepsilon}$ yields

$$b^n M_{s,n}(\varepsilon) \leq Z_{s,n}(\varepsilon) \leq C_s b^n M_{s,n}(\varepsilon). \quad (7.7)$$

Thus the pressure limit exists and equals $\log \lambda_s(\varepsilon)$.

Completion of theorem 2.1. Take the logarithm of equation (7.2). Its quadratic logarithmic correction is $O(\delta^2)$ and is absorbed into $O(\delta^{1+s-\alpha})$. The preceding comparison identifies the result with the pressure in equation (2.7). Compact-parameter uniformity follows from the uniform estimates in the spectral gap, the local integral lemmas, and the strong translation bound. The isolating resolvent circles may be chosen at a fixed relative distance from β_s because the normalized gap is uniform on such compact sets. This proves every assertion of the theorem. $\square$

Proof of corollary 2.3. Since $J_s > 0$ and $\|\varepsilon\|_1 \geq \delta$, the error in equation (2.10) is smaller than $(J_s/2) \|\varepsilon\|_1$ for sufficiently small nonzero ε . This proves the strict local minimum. Substitute $\varepsilon = tv$ and divide by $t > 0$ to obtain equation (2.13). The directional derivative is even in v and positive for nonzero v , so it cannot be a linear functional. Common displacement gives the displayed left and right derivatives and the optimal pointwise exponent. $\square$

8 Finite-size law and explicit amplitude

The contour construction gives a uniform decomposition

$$\beta_s^{-n} \mathcal{L}_{s,\varepsilon}^n = \left(\frac{\lambda_s(\varepsilon)}{\beta_s} \right)^n \Pi_{s,\varepsilon} + R_{n,s,\varepsilon}, \quad \|R_{n,s,\varepsilon}\| \leq C\rho^n \quad (8.1)$$

for some $\rho < 1$. For example, integrate the complementary resolvent on a circle in the normalized spectral gap. Its uniformly bounded resolvent gives the displayed estimate, with a slight enlargement of the spectral radius if necessary.

Proof of theorem 2.4. By equation (7.2),

$$\log \frac{\lambda_s(\varepsilon)}{\beta_s} = J_s \|\varepsilon\|_1 + O(\delta^{1+s-\alpha}).$$

When $n\delta \leq T$, the accumulated logarithmic error is bounded by $C_T \delta^{s-\alpha}$. Both the main exponential and the perturbed exponential are uniformly bounded on this time window. Therefore

$$\left| \left(\frac{\lambda_s(\varepsilon)}{\beta_s} \right)^n - e^{nJ_s \|\varepsilon\|_1} \right| \leq C_T \delta^{s-\alpha}.$$

Combine this with equations (7.3) and (8.1) to obtain equation (2.14).

Apply the operator to 1 and integrate. By equation (7.5), the resulting normalization is $\beta_s^{-n} b^n = b^{sn}$. The beta-integral evaluation

$$\int_0^1 |\sin \pi x|^a dx = \frac{\Gamma((a+1)/2)}{\sqrt{\pi} \Gamma(1+a/2)}, \quad a > -1,$$

and the reflection identities [4] give

$$\left(\int H_s \right) \left(\int H_s^{-1} \right) = \frac{\Gamma((1+s)/2) \Gamma((1-s)/2)}{\pi \Gamma(1+s/2) \Gamma(1-s/2)} \quad (8.2)$$

$$= \frac{2}{\pi s} \tan(\pi s/2) = A_s. \quad (8.3)$$

This proves equation (2.15). With $\varepsilon = h\mathbf{v}$, $n\|\varepsilon\|_1 = nh\|\mathbf{v}\|_1$ and the two error terms tend to zero under the hypotheses of equation (2.16). The same argument uses uniform constants on the stated compact parameter sets. $\square$

An exact relation between the cusp and the amplitude is

$$J_s = \frac{\pi^2 s^2}{2} A_s. \quad (8.4)$$

The factor $b - 1$ in a common translation counts zeros; the amplitude A_s itself is independent of the base. Neither fact extends to an arbitrary changed atomic mask without further analysis.

9 Consequences, examples, and normalization boundaries

9.1 Two concrete values in the newly covered range

At $s = 1/4$, corresponding to pressure order $q = 1/8$ in the binary squared-mask convention,

$$J_{1/4} = \frac{\pi}{4}(\sqrt{2} - 1), \quad P_{2,1/4}(c) = \frac{3}{4} \log 2 + \frac{\pi}{4}(\sqrt{2} - 1)|c| + O_\alpha(|c|^{5/4-\alpha}).$$

At $s = 1/2$, corresponding to $q = 1/4$,

$$J_{1/2} = \frac{\pi}{2}, \quad P_{2,1/2}(c) = \frac{1}{2} \log 2 + \frac{\pi}{2}|c| + O_\alpha(|c|^{3/2-\alpha}).$$

The second point is not a new phase transition. It was only the point where the old norm-square proof ceased to control its remainder.

9.2 Opposite zero motions do not cancel

For $b = 3$, move the zeros at $1/3$ and $2/3$ by t and $-t$. Their mean displacement is zero, but

$$P_{3,s}(t, -t) = (1 - s) \log 3 + 2J_s|t| + O(|t|^{1+s-\alpha}). \quad (9.1)$$

A perturbation rule based only on the center of the zero configuration would incorrectly predict no first-order response. The pressure sees the total absolute opening of the simple zeros.

9.3 Smooth zero trajectories and amplitude changes

Corollary 9.1 (Trajectory law). *Suppose $\varepsilon_j(t) = v_j t + O(t^2)$ as $t \rightarrow 0$. For fixed $0 < s < 1$ and $0 < \alpha < s$,*

$$P_{b,s}(\varepsilon(t)) = (1 - s) \log b + J_s \sum_j |v_j| |t| + O(|t|^{1+s-\alpha}). \quad (9.2)$$

If the mask is additionally multiplied by a positive factor $e^{a(t)}$, its pressure is increased by exactly $sa(t)$.

Proof. The absolute value is Lipschitz, so $\|\varepsilon(t)\|_1 = |t| \sum |v_j| + O(t^2)$; apply theorem 2.1. Multiplication of the mask by $e^{a(t)}$ multiplies every transfer operator by $e^{sa(t)}$, or every n -step product by $e^{nsa(t)}$. Taking logarithmic growth gives the exact second assertion. $\square$

This normalization clause is essential. A varying leading coefficient adds an ordinary signed linear term and can obscure the cusp. The ℓ^1 law is stated for the fixed coefficient modulus in equation (2.1).

9.4 Digital coefficients and the Thue–Morse connection

Write $p_\varepsilon(z) = \sum_{r=0}^{b-1} a_r(\varepsilon) z^r$. If $k = \sum_{j=0}^{n-1} d_j b^j$ with $0 \leq d_j < b$, then

$$[z^k]Q_{n,\varepsilon}(z) = \prod_{j=0}^{n-1} a_{d_j}(\varepsilon).$$

Thus independently moving the zeros produces a natural family of complex digit weights. It does not merely perturb an abstract transfer operator.

For common phase displacement one may choose the equal-modulus polynomial with coefficients $b^{-1}e^{-2\pi i c r}$. Its product is

$$b^{-n} \sum_{k=0}^{b^n-1} e^{-2\pi i c s_b(k)} e^{2\pi i k x},$$

where $s_b(k)$ is the base- b digit sum. Its modulus is the one used throughout this article. For $b = 2, c = 1/2$, the coefficients reduce to $(-1)^{s_2(k)}$, the Thue–Morse signs. At the atomic phase $c = 0$ the polynomial is the normalized Dirichlet polynomial $b^{-n} \sum_{k=0}^{b^n-1} z^k$. The singular limit studied here therefore starts from the simplest digital product but has nontrivial fractional moment response.

9.5 Endpoint information that does not justify exchanging limits

The explicit constants satisfy

$$J_s = \frac{\pi^2 s^2}{2} + \frac{\pi^4 s^4}{24} + O(s^6), \quad A_s = 1 + \frac{\pi^2 s^2}{12} + O(s^4) \quad (s \downarrow 0), \quad (9.3)$$

$$J_s \sim \frac{2}{1-s}, \quad A_s \sim \frac{4}{\pi^2(1-s)} \quad (s \uparrow 1). \quad (9.4)$$

These are expansions of the explicit constants, not uniform joint-limit theorems for pressure. The proof uses a Hölder exponent below s , and the dual density ceases to be integrable at $s = 1$. At criticality the predecessor report finds a different spectral mechanism involving a two-dimensional collision [3]. The present theorem does not replace that critical analysis.

10 Numerical diagnostics and reproducibility

10.1 What was actually run

The package includes `code/verify.py`, its complete output `data/verification.json`, and `data/run.log`. The included run used the `-full` option. Its checks have three independent roles: to evaluate the universal singular integral, to test the signed weak response on a nonconstant function, and to compare finite-dimensional transfer approximations with the pressure and finite-size formulas. They do not form part of the analytic proofs.

The J_s integral was evaluated at 65 decimal digits after substitutions removing the algebraic singularity at zero and mapping the infinite tail to a finite interval. At $s = 0.1, 0.25, 0.5, 0.75, 0.9$ its discrepancy from the elementary expression was below 10^{-30} . This is a high-precision quadrature check, not a rigorous interval enclosure.

10.2 A test that detects the principal-value sign

For the single binary zero $z = 1/2$, use

$$g(x) = 1 + 0.2 \cos(2\pi x) + 0.1 \sin(2\pi x).$$

Then $g(z) = 0.8$ and

$$\text{PV} \int_0^1 \pi \cot \pi(x - 1/2) g(x) dx = -0.1\pi.$$

The sine term follows from $\tan(\pi x) \sin(2\pi x) = 2 \sin^2(\pi x)$; the other terms have zero principal value by reflection. Hence the limits of $|h|^{-1} \int q_{z,\pm h} g$ are

$$0.8J_s \pm 0.1\pi s.$$

The code checks both signs at four exponents and three phase sizes. A check with the constant test alone would miss an incorrect sign in the principal-value part.

The double-precision adaptive quadrature emitted a roundoff convergence warning in this diagnostic family. That warning is retained in the run log. The output includes the quadrature error estimates; it is not silently promoted to a certified result. The integral identity and its sign are established by lemma 5.2, not by this computation.

10.3 Pressure: mesh comparisons

Collocation uses a uniform periodic grid and linear interpolation at each inverse-branch point. The resulting operator is nonnegative. Power iteration is stopped when the sup norm change of the normalized iterate is less than 3×10^{-13} ; the matrix residual is recorded. The coarse and fine meshes below have 262 144 and 524 288 points. The table shows

$$\frac{P_{b,s}^{\text{grid}}(h\mathbf{v}) - (1-s) \log b}{h}.$$

For $b = 2$ the velocity is (1) ; for $b = 3$ it is $(1, -1)$.

b	s	h	coarse grid	fine grid	predicted limit
2	0.20	0.01	0.180110	0.180113	0.204153
2	0.20	0.003	0.194370	0.194378	0.204153
2	0.20	0.001	0.200329	0.200320	0.204153
2	0.25	0.01	0.284731	0.284733	0.325323
2	0.25	0.003	0.308837	0.308843	0.325323
2	0.25	0.001	0.318789	0.318782	0.325323
2	0.50	0.01	1.273868	1.273868	1.570796
2	0.50	0.003	1.446803	1.446805	1.570796
2	0.50	0.001	1.519991	1.519989	1.570796
3	0.30	0.01	0.952195	0.952194	0.960433
3	0.30	0.003	0.958221	0.958219	0.960433
3	0.30	0.001	0.959884	0.959841	0.960433

The low-exponent binary rows approach the predicted coefficient, and the base-three opposite-motion rows distinguish the ℓ^1 law from a mean-displacement rule. Mesh changes remain visible after division by h . A small matrix residual only checks the discretized eigenproblem; it does not bound the distance to the infinite-dimensional spectrum. No number in the table is labeled a certified continuum digit.

10.4 Finite-size amplitude

For $b = 2, s = 1/4, \mathbf{v} = (1)$ and $nh = 0.1$, the prediction is $A_{1/4} e^{0.1J_{1/4}}$. Direct iteration of the normalized collocation operator on the fine mesh gives:

n	h	$b^{sn} M_n$ (grid)	limiting value
25	0.00400	1.0780086	1.0896650
50	0.00200	1.0828660	1.0896650
100	0.00100	1.0857978	1.0896650
200	0.00050	1.0875046	1.0896650

The convergence includes both phase error and finite-size relaxation; it is not expected to be exact at a fixed mesh or a fixed number of steps. The gamma-product and elementary expressions for A_s are also compared independently in the script.

10.5 Rebuilding

From the package root, run

```
python -m pip install -r requirements.txt
OPENBLAS_NUM_THREADS=1 python code/verify.py --full
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

The environment-variable syntax is for a POSIX shell; on Windows it can be set using the shell's equivalent syntax or omitted. A run without `--full` uses smaller meshes. The numerical output records the Python and numerical-library versions. The PDF is built from the provided source and two generated table fragments; all are in the ZIP.

11 Proof boundaries and a formalization route

11.1 Dependency audit

The mathematical dependency chain is short enough to audit explicitly. The pressure theorem requires the atomic spectral gap ([proposition 4.2](#)), the strong–weak perturbation identity ([lemma 3.1](#)), the weak pairing bound and scalar coefficient ([theorem 6.1](#)), and the pressure–moment comparison ([lemma 7.3](#)). The atomic gap follows from an explicit weighted Riemann sum and a finite-iterate Hölder estimate. The weak pairing bound follows from a one-zero scaling calculation and cancellation of its odd tail; independent zero motions are added by a partition of unity with controlled mixed terms. No assumption of strictly positive weights, no unproved linear-response theorem, and no numerical spectral approximation is inserted into this chain.

The special functions only evaluate the universal integral and the amplitude. Leaving those two quantities as their convergent integrals would preserve the qualitative cusp and finite-size statements. The closed forms make the laws explicit.

The article proves a local result relative to a specific family. In particular, it does not prove a general universality theorem for all masks with the same number of zeros. The atomic coboundary and its spectral gap are indispensable model-specific structure. Independent motions are allowed because they start from exactly this atomic mask.

11.2 A staged Lean implementation

No Lean file is included or claimed to compile. A practical future formalization can isolate three layers. The finite layer consists of the root-factor mask identity, the digit-product coefficient formula, and the degree bound $b^n - 1$. The real-analysis layer consists of the one-zero integral estimates, principal-value functionals on Hölder tests, and the weighted quadrature lemma. The operator layer consists of rank-one exponential relaxation, resolvent projections, and [lemma 3.1](#).

The strong-weak eigenvalue lemma is a particularly reusable target: once a suitable spectral-projection API exists, its central equality is just

$$\lambda - \beta - \nu(\Delta h) = \nu\Delta(h_\Delta - h).$$

The essential formal issue is preserving the different norm estimates, not replacing both by one ambient operator norm. For the singular integrals, symmetric cutoff limits should be explicit definitions; otherwise a principal value can be mistaken for an absolutely convergent integral.

A useful executable certificate project would prove rigorous finite-dimensional upper and lower spectral bounds with a separately bounded interpolation error. The existing floating-point verifier is not such a certificate, and a future formal checker should not treat its output as trusted theorem data.

12 Further research questions

Research question 1 (The sharp next term). Can the remainder in equation (2.10) be sharpened to an asymptotic term? Determine whether the next correction is an ordinary quadratic term, a fractional term, or a logarithmic or log-periodic contribution. The present proof bounds the change of the eigenfunction only in a strong norm and does not compute its effect at the moved zeros.

Research question 2 (Removing the arbitrary Hölder loss). For fixed $0 < s < 1$, can one replace $O(\delta^{1+s-\alpha})$ for every $\alpha > 0$ by $O(\delta^{1+s})$, perhaps with a logarithmic factor? A weighted space adapted simultaneously to the atomic point and the mask zeros could avoid choosing a smaller Hölder exponent. The optimal total-variation law equation (6.6) is a constraint on any such argument, not by itself an answer.

Research question 3 (A joint limit as the moment order tends to zero). The coefficient behaves as $J_s \sim \pi^2 s^2/2$, but the proof is not uniform as $s \downarrow 0$. Determine the joint pressure asymptotic when $s \log(1/\delta)$ remains bounded. Does the apparent quadratic small-order response retain a universal scaling function, or do additional orbit effects become visible?

Research question 4 (Independent zero motions at criticality). At $s = 1$, does the critical splitting depend only on $\sum_j |\varepsilon_j|$, or on a different weighted combination of zero motions? The atomic left density is no longer integrable and the rank-one argument here fails. One must compute the perturbation of the critical spectral block rather than substitute $s = 1$ in the subcritical coefficient.

Research question 5 (Uniform matching as $s \uparrow 1$). Can the compact-subcritical theorem be matched quantitatively to the critical two-parameter law in [3]? An overlap region must control the blowup of ν_s and of the reduced resolvent. The pole $J_s \sim 2/(1-s)$ identifies what must be matched but does not justify the interchange of limits.

Research question 6 (Two-point regularity and nonzero phases). Is the common-phase pressure locally Lipschitz on a whole neighborhood of the atomic phase for every fixed $0 < s < 1$? At nonzero phases, what regularity is forced or obstructed by returns of singularities under T_b ? The estimate proved here is anchored to the atomic left functional and does not directly give a bound between two arbitrary nearby nonzero phases.

Research question 7 (Higher-multiplicity zeros). For an atomic model with zero multiplicities $r_j > 1$, what replaces the integrability threshold and the coefficient J_s ? A local factor suggests replacing s by $r_j s$, but a pressure theorem also needs an actual spectral gap and compatible dual density. Determine precise hypotheses under which local multiplicities determine a tangent law.

Research question 8 (Response of eigenfunctions and equilibrium data). Find a distributional or weighted-space expansion of the right and left spectral vectors themselves. This could yield first-order laws for observables and, in the probability-normalized subfamily, equilibrium measures. The pressure theorem only uses a norm estimate for the perturbed eigenfunction and does not identify its singular profile.

Research question 9 (Certified computation near a cusp). Develop interval quadrature and a proof-producing spectral enclosure whose error remains smaller than the $O(\delta)$ pressure increment. Uniform collocation spends many points away from the few translated zeros. A decomposition that treats their singular parts analytically could improve both efficiency and the trust boundary.

Research question 10 (An abstract moving-singularity response theorem). Which expanding maps and finite-zero weight families admit a weak expansion of the form “Dirac masses plus principal values” together with a strong-weak eigenvalue remainder? Formulate a theorem with explicit assumptions on the atomic spectral data, zero separation, local exponents, and test-function regularity. The digital masks here provide a complete worked example, not a proof for all such systems.

13 Conclusion

The lower cutoff in the previous subcritical argument was a limitation of its perturbation estimate, not a spectral transition. Pairing the perturbation with the atomic left eigenfunctional preserves the cancellation around translated zeros and makes the nonlinear remainder smaller than the cusp for every $0 < s < 1$. The resulting theorem closes the repository’s explicitly remaining subcritical interval, supplies a quantitative error, and extends the answer to independent zero motions. In this larger parameter space the first pressure increment is $J_s \|\varepsilon\|_1$, while finite products have the scaling amplitude A_s and time scale $n \|\varepsilon\|_\infty$. The unresolved questions now concern sharper corrections, endpoint matching, nonzero phases, and certified implementation, rather than whether the leading fixed-subcritical cusp exists.

References

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